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Innovative Mechanical Solutions for Scale Removal: Optimizing Pre Plug and Abandon Operation Through Robotic Milling Solution to Achieve Multiple Milling OD's in One Run

Matthew Warren Powell, Eyad Rai, Zaher Taki, and Arild Flokenes, Welltec, Bergen, Vestland, Norway

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Abstract

Milling and wellbore cleanout operations are fundamental to successful well intervention, particularly in aging wells nearing the plug and abandonment (P&A) phase. These interventions are often challenged by scale build-up, tubing restrictions, and the need to access specific depths to install permanent barriers. Conventionally, such operations require multiple runs with separate cleanout assemblies—ranging from milling, explosives, and caliper logging runs—to first gain access and then ensure the wellbore is clean enough to deploy and seal a plug. Each additional run adds complexity, cost, rig time, and environmental footprint, making single-run solutions highly desirable.

To address these inefficiencies, a single-run multi-OD milling tool was developed with the purpose of enabling operators to bypass restrictions, mill scale, and clean the tubing to drift ID—all within one continuous descent. This tool was designed through a close collaboration between field operations, design engineering, and sales teams to meet an urgent market need: a cost-effective, emissions-reducing, and rig-time-saving cleanout solution that could adapt to variable wellbore geometries and scale conditions.

The tool combines a fixed lead bit with a second-stage expandable milling bit. The lead bit allows the bottom hole assembly to pass through scale-narrowed or ID restricting tubing, while the expandable bit is selectively activated once at the target depth. Surface-controlled activation enables precision depth targeting and real-time control of the milling passes.

The first commercial deployment of this tool occurred in a North Sea well with known caliper-logged restrictions and heavy scale buildup. The planned operation required passing through an upper ID restriction to reach the intended plug setting depth, followed by cleaning the interval to near-drift ID to ensure effective plug sealing. Due to the sensitivity of the plug to debris and scale, high cleaning efficiency and accuracy were crucial for success. The full BHA—including a power mechanical tractor, stroker, and the single-run multi-OD milling tool—was mobilized and deployed in a single descent.

The lead mill bit successfully cleared the scale restriction, allowing further descent to the scale-heavy interval. Once at the plug setting depth range, the expandable milling bit was activated and engaged in four milling passes. High and variable current readings during the first pass confirmed active engagement with scale, while diminishing current feedback on subsequent passes suggested effective cleaning. Following

milling, the BHA was retrieved, and a multi-finger caliper log was run to verify internal diameter restoration. The data confirmed complete scale removal in the plug setting zone. The plug was then run to depth, set, and pressure tested with success, allowing the P&A sequence to continue.

This operation validated the tool's design and confirmed the success of the single-run approach. Notably, the lessons learned included the realization that current monitoring during activation provides sufficient feedback to verify scale removal—eliminating the need for caliper logs in future operations. Historical operations attempting similar objectives have required as many as seven separate runs over two days; this new approach achieved the same goal in a single run, reducing rig time by more than 24 hours and delivering substantial cost and emissions savings. Additionally, tool performance helped validate new arm designs and highlighted the potential of different material configurations for improved milling efficiency across a range of scale conditions.

This paper outlines the full development process, including design iterations and bench testing, and describes the operational execution in detail. It shares comprehensive results and key takeaways from the deployment. The introduction of the single-run multi-OD milling tool marks a significant step forward in efficient wellbore cleanout and intervention methodology. It not only simplifies operations but also reduces environmental impact and total cost, paving the way for broader implementation in complex well interventions and P&A projects globally.

Keywords: Well Intervention, Plug and Abandonment (P&A), Scale Removal, Milling Operations, Expandable Bit, Single-Run Operation, E-line Conveyance, Wellbore Cleanout, Multi-OD Milling, Internal Diameter (ID) Restoration, Tubing Drift, Downhole Tooling, Power Mechanical Tractor, Power Mechanical Stroker, Section Milling, Sonic Pulse Technology, Explosive Percussion Cleaning, Operational Efficiency, Rig Time Reduction, Barrier Placement

Introduction

Milling operations takes a major part in well interventions across the oil and gas industry. Whether used to remove scale, eliminate restrictions, or gain access to deeper intervals in the wellbore, mechanical milling has long served as a foundational technique for enabling plug placement, abandonment preparation, and overall well access. In Pre-Plug and Abandonment operations, the ability to reach the intended setting depth and establish a clean, drift-confirmed internal diameter is critical for the successful deployment and pressure testing of permanent barrier elements. Certain plug types are highly sensitive to residual scale and require tubing IDs to be restored as close as possible to nominal tolerances to ensure sealing integrity and valid pressure testing.

Conventional milling approaches typically utilize a fixed-diameter lead bit, which creates a uniform bore down to depth. While this method can provide access for follow-on intervention tools, its effectiveness is limited to scenarios where the entire well section can be milled with a constant diameter. In more complex wells—where internal diameters vary due to completion components or where heavy scale is present—traditional methods often fail to reach target depths or fully restore the tubing ID. These techniques are commonly conveyed via mechanical wireline tractors with e-line milling tools or through coiled tubing techniques, depending on operational requirements.

To address more severe buildup or obstructions, alternative technologies such as sonic pulse-based treatments and explosive percussion-scale removal have been adopted in industry. Although these methods can offer partial restriction removal in localized zones, their effectiveness is highly dependent on well conditions. Moreover, they introduce operational complexity, require specialized permitting and surface equipment, and often do not guarantee consistent ID restoration. Post-treatment acid washes and mechanical verification runs—such as caliper logging—are frequently required, increasing both rig time and intervention costs.

In recent North Sea campaigns, operators observed that these multi-step intervention strategies were significantly increasing rig time, emissions, and overall cost. Jobs that once required as many as seven individual runs—including sonic treatments, explosive cleaning, milling, and verification—prompted the search for a more consolidated and mechanical solution. This need led to the development and deployment of a single-run multi-OD milling intervention tool capable of addressing access and drift restoration in one integrated operation. The tool's ability to first pass through restricted IDs and then expand to mill a larger section introduces an effective bit-to-scale solution in downhole cleanout capabilities.

This paper details the methodology, execution, and field performance of this intervention in a challenging Pre-P&A scenario. The case highlights how a single-run mechanical milling solution can reduce complexity, improve operational efficiency, and contribute to significant reductions in rig time, CO₂ emissions, and cost.

Solution Development/Technology Overview

The development of the single-run multi-OD milling tool emerged from a clear operational need: to combine proven milling methods with selective, expandable section milling functionality in a single, versatile platform. While conventional milling tools with fixed-diameter lead bits had long demonstrated reliable performance for creating access downhole, they lacked the flexibility to address variable-ID sections or restore full tubing drift dimensions deeper in the well. Separately, a single-phase expandable milling system had existed—where an internal pressure command would open an expanded cutting bit at depth—but no solution existed that could integrate both capabilities in a controllable, sequential manner.

To address this gap, the solution was engineered as a dual-pressure, surface-controlled milling tool capable of performing both functions in a single run.

The tool design consists of:

- A lead milling bit, used to gain wellbore access. This lead bit can be selected from various styles and sizes of off-the-shelf designs, including scale milling bits, elliptical bits, or core bits, allowing for high operational flexibility.
- Expandable bit. This bit remains deactivated during conveyance and lead milling operations and is activated by a second command delivered from surface. This dual control allows the operator to run either the lead bit or the expandable bit independently, providing unmatched flexibility and precision.

To enhance its adaptability across different well conditions, the expandable bit can be customized with different types of milling elements to tackle different type of scale. This configuration gives the operator the ability to tailor the downhole cutting performance based on the scale type and severity.

The initial deployment of the tool was designed to address a specific Pre-P&A challenge in a well with a restricted 5.5" tubing section above a scaled 7" interval. The objective was to mill a lead path through scale using a 4.5" lead bit, then expand to achieve a 6.2" internal drift at the plug setting depth. To meet this requirement, a tool was designed with a 4.2" to 6.2" expansion range, enabling it to be conveyed through restrictions and then restore full drift at the target zone. The expandable section was intentionally designed with a smaller body OD than the lead bit to ensure efficient pass-through of tight ID profiles before activation.

As demand for this technology grows, further development efforts are focused on expanding the available bit configurations and OD ranges. These improvements will allow the system to be deployed in a wider variety of tubing sizes and intervention scenarios, reinforcing its role as a highly adaptable solution for downhole milling and plug preparation operations.

Operational Execution & Results

Prior to the operation, the well was evaluated using historical multi-finger caliper data which confirmed extensive scale buildup in the 5.5" and along the 7" tubing. Several intervals displayed inner diameter (ID) restrictions, and a particularly severe accumulation was identified near 2002.5 meters. The planned objective was to prepare a clean bore suitable for the deployment of an expandable plug, which required a drift-verified ID to achieve proper sealing integrity. Due to the plug's sensitivity to scale and surface roughness, a near-original tubing ID was necessary to ensure successful placement and pressure integrity testing.

To address this challenge, a toolstring comprising a power mechanical tractor, a power mechanical stroker, and the single-run multi-OD milling tool was mobilized offshore. The combination of tools enabled controlled conveyance, targeted force application, and selective milling functionality.

The operation commenced by deploying the BHA to the expected hold-up depth (HUD) at 1998.5 meters. At this point, the lead bit of the milling tool was activated to mechanically remove the high-density scale buildup. The lead bit milled down to 2002.5 meters, clearing the most restrictive section and enabling access to the plug setting depth.

With access established, the BHA was tractored further downhole to the target zone, reaching 2081 meters. The planned plug setting interval ranged from 2081 to 2087 meters, requiring complete ID restoration to allow plug sealing and anchoring. At this depth, the expandable milling bit was deployed, and four full milling passes were executed to ensure thorough cleaning.

During the first pass, tool surface read-out feedback indicated elevated and sporadic readings indicating active milling and bit engagement with residual scale. The subsequent second, third, and fourth passes exhibited decreased and more stable profiles, suggesting progressive scale removal and reduced mechanical resistance. These trends served as downhole verification of successful milling.

Following the milling operation, the BHA was retrieved, and a multi-finger caliper log was run across the milled interval. The caliper data confirmed complete scale removal throughout the plug setting zone, validating the mechanical cleaning achieved by the expandable bit, [Figure 1](#) and [Figure 2](#).

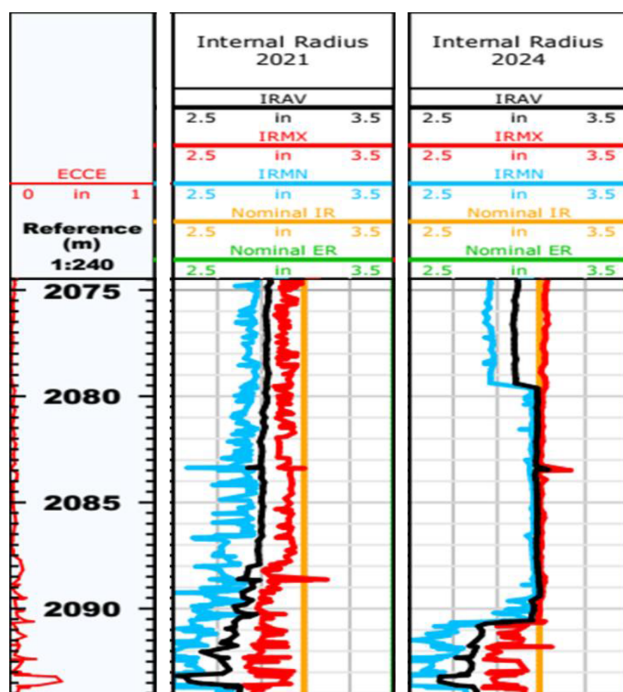


Figure 1—Multi-finger caliper log illustrating the before and after milling affect. Left side scale shows the radius tubing & scale radius prior to milling, and the right side shows the radius after the expandable bit milling.

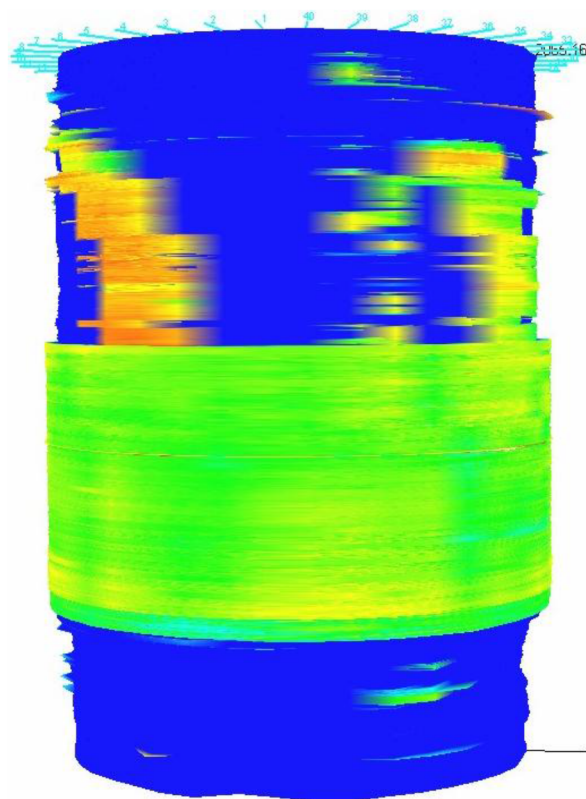


Figure 2—3D rendering of the post milling caliper data showing the clean setting area for the plug.

The plug was then successfully conveyed, set, and pressure-tested without issue, meeting all operational objectives and allowing the P&A program to continue as planned.

This job marked the first field deployment of the single-run multi-OD milling intervention solution. The results demonstrated the viability of the technology in delivering a fully integrated, single-run solution capable of navigating restrictions, opening access, and performing localized section milling. A key operational learning was that three to four milling passes, combined with real-time current monitoring, proved sufficient to verify scale removal—potentially eliminating the need for additional verification runs in future deployments.

Lessons Learned

The successful deployment of the single-run multi-OD milling intervention solution yielded several key insights with respect to operational efficiency, tool development, and post-milling verification:

Significant Time and Cost Savings

This solution consolidated what has traditionally been a multi-step, multi-run process into a single, continuous operation. Comparable interventions have required up to seven separate runs over two days to achieve similar objectives—particularly when using sonic pulse treatments and explosive scale removal methods followed by verification passes. By contrast, this single-run power mechanical intervention approach eliminated the need for multiple runs, thereby reducing rig time, HSE exposure, and logistical complexity. In high-cost offshore environments, this translates into millions of dollars saved per campaign, representing a significant improvement in cost-efficiency and project economics.

Expandable bit Optimization and Development

The trial also accelerated development around expandable milling bit technology:

- **Material Variation:** To provide operational flexibility across scale and metal removal applications.
- **Performance Tuning:** Field data has helped optimize bit geometry and control to improve milling efficiency and durability.
- **Range Adaptation:** The success of the initial 4.2"-to-6.2"-bit range has prompted design extensions to accommodate a broader spectrum of tubing IDs and completion types.

Three Milling Passes Are Sufficient

The field execution validated that three to four milling passes with the expandable bit were sufficient to achieve full ID restoration. This provides a clear guideline for future interventions using this tool, streamlining planning and execution without compromising effectiveness.

Conclusion

Wellbore scale and internal diameter restrictions remain some of the most persistent challenges in well intervention, particularly in mature assets. Historically, operators have relied on multi-run intervention programs involving conventional milling options, dedicated cleanout tools, and separate verification runs to achieve the dual objectives of clearing restrictions and restoring tubing ID to drift dimensions. These approaches, while technically viable, carry significant inefficiencies: they increase rig time, elevate operational risk, drive up cost, and expand the carbon footprint of intervention campaigns.

The introduction of the single-run multi-OD milling tool represents a transformative advance in addressing these longstanding challenges. Developed with an integrated design philosophy, the tool combines a conventional fixed-diameter lead mill with a second-stage expandable milling system that can be selectively activated from surface. This dual functionality allows operators to pass restrictions initially and subsequently expand the tubing ID at the precise setting depth—all within one tool run. The expandable section has been engineered with multiple arm and bit configurations to accommodate different scale compositions and tubing geometries, enhancing operational flexibility across various well conditions.

The first commercial deployment demonstrated the full range of this tool's capabilities. The power mechanical tractor and stoker assembly successfully conveyed the milling bottom hole assembly to the hold-up depth (HUD). The lead mill was engaged to create access to the plug setting depth. Once positioned, the expandable section was activated, and four controlled passes were performed across the plug setting interval. Real-time surface current measurements clearly indicated initial aggressive milling activity followed by progressively smoother passes, confirming removal of scale and validating the effectiveness of the cleaning process. Post-job caliper logging confirmed a fully restored drift ID and a clean setting zone. The subsequent plug installation was executed without complications, and a successful pressure test validated isolation integrity.

The lessons learned during this operation further strengthen the value proposition of the technology. Real-time monitoring of current proved to be a reliable indicator of milling progress, demonstrating that caliper verification runs—previously considered mandatory—may no longer be required. This simplification not only reduces cost and rig time but also minimizes operational exposure. Additionally, the test provided insight into the optimal number of expandable passes, with three to four passes proving sufficient for complete cleaning across the setting interval.

The impact of this technology extends beyond a single operation. Conventional methods had previously required up to seven separate runs and more than two days of rig time to achieve similar outcomes. The single-run multi-OD milling tool reduced this to a single, streamlined intervention. This translates directly into millions of dollars in potential savings per campaign while simultaneously reducing the carbon emissions associated with extended rig operations. The integration of multiple functions into one tool embodies a more sustainable and efficient intervention methodology, aligning with the industry's broader objectives of lowering environmental impact while maintaining operational excellence.

In conclusion, the single-run multi-OD milling tool is not only an incremental improvement on existing milling practices but a step-change in how scale cleanout and tubing restoration are performed. By combining restriction removal and ID restoration in one intervention, supported by real-time validation, the tool offers operators a faster, safer, and more cost-effective pathway to achieving well integrity objectives. Its successful deployment in the North Sea underscores its readiness for broader application, and with continued development of expandable arm designs, the technology holds the potential to redefine the standard for milling and cleanout operations across the global industry.

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Finally, we would like to thank the sales team for recognizing the business need within the industry and driving the development of this technology. Their close collaboration with clients and understanding of operational pain points laid the foundation for this successful deployment and its future potential.

Symbols

Symbol	Description	Units
D	Depth	m
ID	Internal Diameter	in / mm
L	Tool length	ft
OD	Outer Diameter	in / mm
P	Pressure	psi / bar
T	Temperature	°F / °C
Tc	Tensile strength	lbs

Nomenclature

Term	Definition
BHA	Bottom Hole Assembly – the toolstring that includes milling and intervention tools.
Caliper	A downhole logging tool used to measure the internal diameter of the wellbore or tubing.
CO ₂ Emissions	Carbon dioxide released during operations, typically related to rig and equipment usage.
Completion	The assembly of downhole tubulars, packers, and other equipment required to produce oil or gas.
Drift Diameter	The verified internal diameter that a tool or plug must pass through to be considered acceptable.
E-Line	Electric line used to convey tools into the wellbore and supply power and telemetry.
Expandable Bit	A milling component that extends at depth to enlarge the milled diameter.
Explosive Percussion	A method of scale removal using controlled explosive charges to break up obstructions.
Fluid Circulation	The movement of fluid through the wellbore to carry debris to surface during milling operations.

HUD	Hold-Up Depth – the depth at which tool progress is initially stopped due to obstruction.
ID	Internal Diameter – the inner measurement of a tubing or casing.
Lead Bit	The primary milling bit used to initiate the milling operation and establish the bore path.
Multi-Finger Caliper	A tool with multiple arms used to capture detailed ID measurements after a milling operation.
Multi-OD Milling Tool	A milling tool capable of operating with different outer diameters in a single run.
P&A	Plug and Abandonment – the process of safely closing and sealing a well at the end of its life.
Power Mechanical Stroker	A surface-controlled tool that applies linear axial force downhole, typically for setting or shifting.
Power Mechanical Tractor	A motorized downhole tool used to pull or push intervention tools to depth within the well.
Scale	Mineral buildup inside the wellbore that restricts flow or access.
Single-Run	A job or operation performed in a single descent into the well, without the need for retrieval and re-run.
Sonic Pulse Technology	A scale removal method that uses high-energy acoustic waves to dislodge debris.
Standoff Component	A part of the toolstring designed to center or space the tool inside the wellbore.
Surface Monitoring	Real-time monitoring of tool parameters and performance from surface during downhole operations.
Toolstring	The full set of tools run together into the well for a specific operation.
Tubing	A pipe inside the well used to produce fluids or perform interventions.

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Appendix 1 – Sample Bottom Hole Assembly

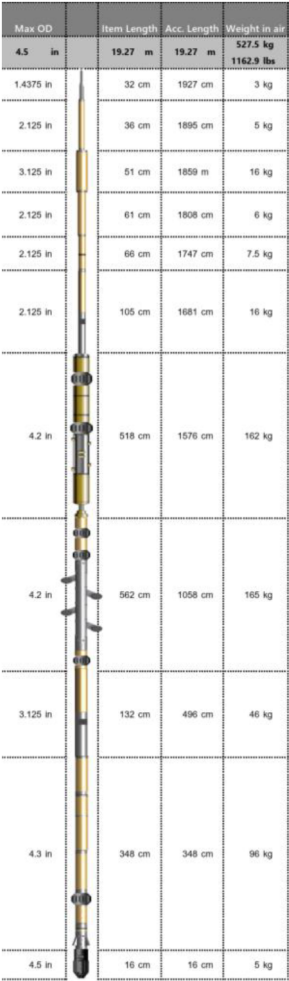


Figure 3—Sample bottom hole assembly diagram for the single-run milling solution.

Appendix 2 – Single-run multi-OD milling tool specifications sheet

Specifications*	Imperial	Metric
• Length	• 12.7 ft	• 3.9 m
• Running OD nominal	• 3 1/8" *	• 80 mm *
• Weight in air	• 190 lbs	• 86 Kg
• Minimum completion ID	• 3.25" *	• 82.5 mm*
• Maximum Bit OD	• 5.0" – 6.2" **	• 127 – 157.5 mm **
• Maximum well pressure	• 25,000 psi	• 1,720 bar
• Maximum well temperature	• 302 F	• 150 °C
• Tensile strength	• 42,000 lbs	• 19,000 Kg
• Compressive strength	• 50,000 lbs	• 22,700 Kg

* Dependent upon configuration.

** Customisation is available

Figure 4—Specifications sheet for the single-run multi-OD milling tool.